

Pursuing a **Minor degree in Artificial Intelligence and Data Science** at CMInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured an **All India Rank of 635** in **Joint Entrance Exam (JEE) Advanced** among 0.15M+ candidates [Oct'21]
- Secured an **All India Rank of 1323** in **Joint Entrance Exam (JEE) Mains** among 1M+ candidates [Sep'21]
- Recipient of the Kishore Vaigyanik Protsahan Yojana (**KVPY**) **Fellowship**, awarded by **IISc Bangalore** [Jan'21]
- Awarded **Best Project** in **Electronics Device Lab** for EEG measuring device, out of **200+** students [Apr'24]

PROFESSIONAL & RESEARCH EXPERIENCE

AI Agent Intern | Attentions.AI

[Sep'24 - Jan'25]

Awarded a Letter of Recommendation in recognition of exceptional technical contributions

- Developed a scalable automation framework using **PyAutoGUI** and **OmniParser** for dynamic web UI interaction
- Leveraged **Phi-3 Instruct** with **few-shot prompting** to generate dynamic action plans for complex workflows
- Achieved **80%** task completion accuracy by integrating **InternVL2 VLM** for visual state validation with retry logic

Text-to-Speech Development Intern | Attentions.AI

[May'24 - Jul'24]

To improve speech naturalness and real-time efficiency in Text-to-Speech systems

- Benchmarked TTS models and optimized the pipeline with **text chunking**, reducing inference latency by **50%**
- Enhanced speech expressiveness via **emotion-conditioned attention** mechanism, improving MOS by over **10%**

Multi Document Multi Modal Reasoning | Thesis Project

[Jan'25 - Present]

Dual Degree Project-1 | Guide: Prof. Ganesh Ramakrishnan, Department of Computer Science and Engineering

- Improved large-scale image retrieval via **submodular subset selection**, boosting performance in 10k+ corpora
- Enhanced query robustness via **data augmentation & context similarity optimization**, improving retrieval by 5%
- Extending to multi-document, 1k+ files using **ensemble-similarity**, LLM-filtering, and Inverted file indexing

Singer Classification using Gesture Features | Research Project

[Aug'24 - Nov'24]

Supervised Research Exposition | Guide: Prof. Preeti Rao, Department of Electrical Engineering

- Implemented a multi-class classification model for Hindustani singers with **88.28%** accuracy using gesture data
- Extracted and analyzed features like **LDJ**, **SPARC** from 3D motion capture data to quantify motion smoothness
- Built an end-to-end pipeline for processing and classifying time-series gesture data using **SVM** with **RBF** kernel

KEY PROJECTS

AI-Generated Image Classification and Artifact Detection | Industrial Project

[Nov'24 - Dec'24]

Inter-IIT Tech Meet 13.0 | Adobe Research Challenge

- Designed pipeline with **DenseNet121** & **MobileNetV3** for AI-vs-Real image classification, achieving **98%** accuracy
- Fine-tuned Qwen2-VL with **QLoRA** and **Tree of Thoughts** to detect 70+ artifacts, achieving an F1-score of **0.79**
- Incorporated **adversarial defenses** (FGSM, PGD), enabling cross-model generalization and CPU inference of **<2s**

IITB Q&A Smartbot | Course Project

[Sep'23 - Nov'23]

EE782 - Advanced Topics in Machine Learning | Guide: Prof. Amit Sethi

- Scraped **20+** IITB websites using **BeautifulSoup** and Requests to build a domain-specific knowledge database
- Built **Agentic RAG** chatbot using **LangGraph** and **ChromaDB** storage with dynamic routing for accurate response
- Integrated **LLaMA2-13B-GPTQ** model with **LangSmith** for real-time evaluation, output moderation, and tracing

Stock Price Prediction using LSTM | Course Project

[Aug'23 - Sep'23]

EE782 - Advanced Topics in Machine Learning | Guide: Prof. Amit Sethi

- Developed a **PyTorch LSTM** module along with a flexible data loader to enable dynamic stock price prediction
- Designed a trading strategy class with **EMA indicator** to automate stock trades based on LSTM's predictions
- Achieved a **21%** profit with LSTM's predictions on test data compared to **12%** profit from buy-and-hold strategy

Denoising of Synthetically Noised Images | Course Project

[Jan'24 - Apr'24]

GNR638 - Remote Sensing and Machine Learning | Guide: Prof. Biplab Banerjee

- Generated blurred image pairs using **3 Gaussian filters** to create diverse synthetic data for robust model training
- Trained a **GAN-based** deblurring model with **VGG perceptual** and **Wasserstein loss**, boosting PSNR by **+5.8 dB**
- Achieved an average **25.69 dB** PSNR on synthetic set using a **ResNet-based generator** for sharp image restoration

OTHER PROJECTS

Image Captioning | Summer Project

[May'23 - Jul'23]

Seasons of Code | Web and Coding Club, IIT Bombay

- Implemented an algorithm on Neural Image Caption Generator by Google using **CNN** as the image descriptor
- Used pretrained **VGGNet16** for feature extraction and built a preprocessing pipeline on the Flickr8k dataset
- Predicted caption sequences with **LSTM**, addressing vanishing gradients and achieving a **BLEU-4 score** of **0.1436**

Developing Face Recognition and Generation Models | Course Project

[Sep'23 - Oct'23]

EE782 - Advanced Topics in Machine Learning | Guide: Prof. Amit Sethi

- Constructed **Siamese network** using **ResNet18**, exploring diverse configurations of dropout, LRs, and optimizers
- Built a validation set for predicting whether a pair of images belong to the same person, achieving **75%** accuracy
- Implemented **DC-GAN** architecture with **BCE** loss, leveraging the LFW dataset to generate synthetic face images

Limestone Data Challenge | Algorithmic Trading Hackathon

[Mar'23 - Apr'23]

Organised by Finance & Analytics Club, IIT Bombay in association with TOWER Research Capital

- Clustered **100 stocks** into **4** distinct sectors using correlation based analysis of returns between stocks and indices
- Predicted sector index returns with **regression models**, achieving up to **R = 0.49 correlation** with the actual prices
- Designed a covariance driven **long-short trading strategy** with net-zero exposure, achieving a **Sharpe ratio** of **1.7**

Bitcoin Trading Strategy | Summer Project

[Jun'25 - Jul'25]

Summer of Quant | Quant Community, IIT Bombay

- Completed a program on Financial Modeling and Time Series Analysis, covering **Derivative Pricing** and Hedging
- Designed strategies with momentum and **breakout reversal** logic, using long-short entries & trailing-stop exits
- Achieved Sharpe ratios up to **8.21**, net profits up to **5.5x**, controlled drawdowns, verified against **lookahead bias**

Understanding Trading of Currencies and Commodities in Stock Market

[Jun'22 - Aug'22]

Finsearch | Finance Club, IIT Bombay

- Secured a place in the **Top 15** out of 240+ teams in a 4-member team, analyzing the **Forex** and **commodity markets**
- Researched major factors influencing short/long-term market trends, formulating data-driven **trading strategies**
- Developed and backtested a model for commodity and Forex trading to validate the performance and accuracy

Detecting Depression through Tweets | Self Project

[Dec'22 - Jan'23]

Detecting signs of depression from tweets using sentiment analysis and deep learning

- Preprocessed tweet data using tokenization, stopword removal, stemming, lemmatization for sentiment analysis
- Developed a two-layer **LSTM** for sentiment prediction, achieving **82%** accuracy vs 50% with **Logistic Regression**

POSITION OF RESPONSIBILITY

WiDS Project Mentor | Analytics Club, IIT Bombay

[Dec'24 - Jan'25]

- Mentored **10** students on Bokeh effect, applying **DeepLab** for segmentation and **OpenCV** for background blur
- Curated a project after exploring multiple ideas to enhance learning in object segmentation and image processing

Teaching Assistant | EE, IIT Bombay

[Aug'25 - Present]

EE353 - Introduction to Data Science and Machine Learning | Guide: Prof. Jayakrishnan Nair

- Coordinating in a group of **7** TAs; responsible for the assessments, grading and counseling of **100+** course students
- Assisting professor in moderating classes, proctoring examinations, catering to students' course-related queries

TECHNICAL SKILLS

- Programming Languages:** C++, Python, SQL, JavaScript, VHDL, Embedded C, HTML, CSS, $\text{\LaTeX}$
- Machine Learning:** NumPy, Pandas, TensorFlow, Keras, PyTorch, OpenCV, Scikit-Learn, NLTK
- Software Tools:** GitHub, Matlab, Selenium, Excel, ArduinoIDE, Intel Quartus, GNU-Radio

COURSES UNDERTAKEN

Data Science and AI

Computer Programming and Utilization, Advanced Topics in AI Data and Policy, Advanced Topics in Machine Learning, Image Processing, Optimization in ML

Maths and Physics

Applied Linear Algebra, Complex Analysis, Differential Equations, Quantum Physics

Electrical Engineering

Probability and Random Processes, Digital Signal Processing, Communication Systems, Markov Chains, Number Theory and Cryptography, Control Systems, Processor Design

EXTRACURRICULAR ACTIVITIES

Technical	Successfully completed 8-week Data Structures and Algorithms Bootcamp by GFG	(2025)
	Mentored 2 teams of TinkerKar by TL, IITB to build an autonomous Line Follower Bot	(2023)
	Completed Game Theory and Tinkering Bootcamp under Learner Space, IIT Bombay	(2022)
	Worked in a team to build an RC trainer aircraft organised by Aeromodelling Club	(2022)
Cultural and Sports	Stood 1st in Gyrations , Inter-Hostel Dance General Championship of IIT Bombay	(2024)
	Participated in Annual InSync Dance Show with footfall of 1.5k+ , by InSync Club	(2022)
	Participated in Intra-Department Badminton Tournament and reached the top-8 teams	(2022)